Decision Tree (ID3 - Classification)

i) Start.

ii) Input training dataset with features X and class labels Y.

iii) If all samples belong to the same class:

- Create a leaf node with that class.
- Stop.

iv) If no features remain:

- Create a leaf node with majority class.
- Stop

v) For each feature, compute Information Gain:

$$IG \leftarrow Entropy(parent) - \sum \frac{|subset|}{|parent|} Entropy(subset)$$

vi) Select the feature with the highest Information Gain.

vii) Split the dataset based on the selected feature.

viii) Repeat recursively steps 3-7 for each subset.

ix) Stop when:
- All nodes are pure, or
- No further splitting is possible.

x) Output the constructed decision tree.

xi) End.